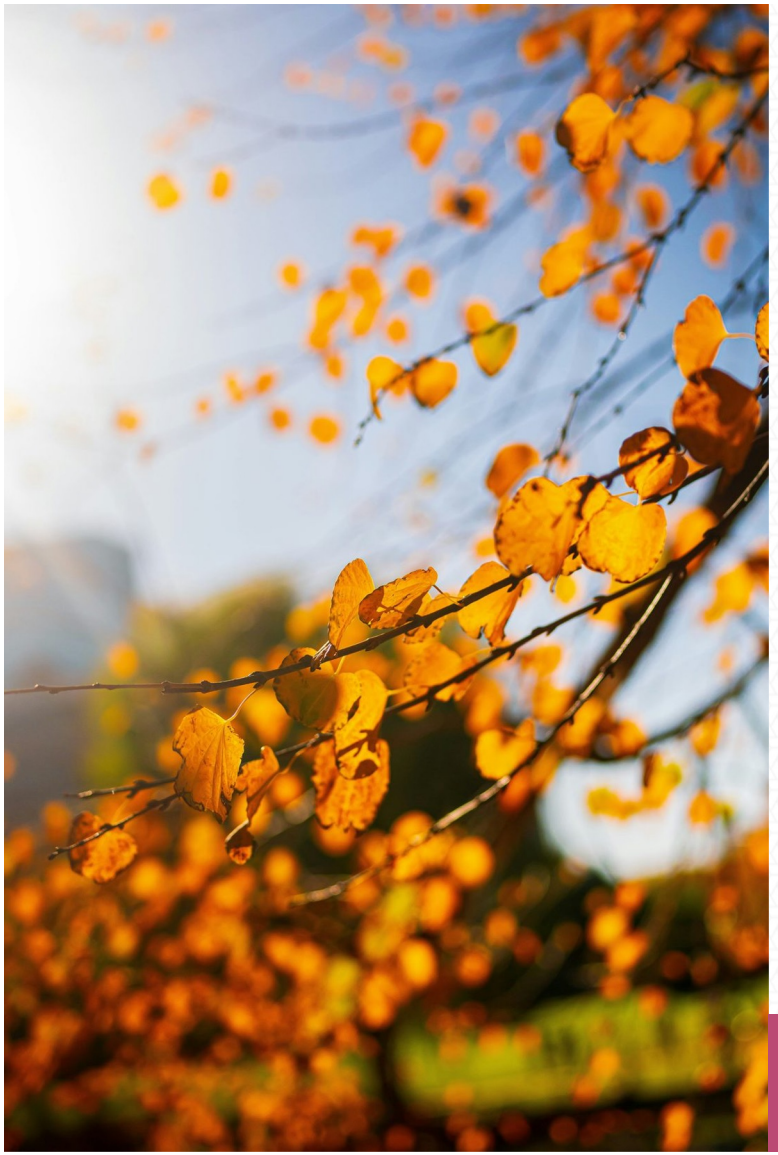


VIS2026

Lossy Compression for Scientific Data

Principles, Tools, and Implications for Visualization

Franck Cappello (ANL), Peter Lindstrom (LLNL),
Sheng Di (ANL), Robert Underwood (ANL),
and Hanqi Guo (OSU)

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Presenters



Franck Cappello, ANL



Peter G. Lindstrom LLNL



Sheng Di, ANL



Robert Underwood, ANL



Hanqi Guo, OSU

Lossy compressors: SZ, ZFP, TTRESH, SPERR, MGARD, LibPressio

Error Analytics: Z-checker, SDRBench, FZ-VIS

ECP projects: SZ, ZFP, CODAR

DOE/NSF: ZF, FZ, etc.

Goals and organization

- What you will learn:
  - Basics as well as the state-of-the-art techniques to reduce scientific data set using lossless and lossy compression
  - Understand the compressibility of scientific datasets, techniques used in compressors for scientific datasets
  - How the most advanced compressors for scientific dataset work
  - How to assess lossy compression errors and distortions
- Please interrupt us for any questions

Outline

Introduction of the Tutorial (5 min.) FC

Section 1: How lossy compression for scientific data works (30 min.) PL

Decorrelation, Approximation (quantization), Coding.

Section 2: Why lossy compression for scientific data and use-cases (20 min.) FC

Motivation, Use-cases, General principles, the role of Shannon Entropy, Limitations of lossless compression, Why scientific data needs specialized lossy reduction

Section 3: Assessing Error in Lossy Compression (20 min.) FC

Beyond PSNR and SSIM: error autocorrelation, derivative preservation, and physically motivated metrics. Progressive compression and level-of-detail rendering for interactive exploration.

Section 4: Feature and Topology Preserving Compression (20 min.) HG

Distortions in volume rendering, isosurfaces, and streamlines. Novel feature-preserving compressors: TopoSZ (contour trees), MSz & pMSz (Morse-Smale segmentations), vector-field critical points, and spectrum-preserving FFCz.

BREAK (30 min)

Outline

Section 5: State of the Art Compressors (60 min.) PL, SD, HG

Understand how state of the art compressors including SZ, ZFP, and Neural Representations are function and are used for scientific computing applications

Section 6: Hands on with useful compression tools (20 min.) RU, HG

How to use the FZ modular compression framework via LibPressio both from a programmatic API perspective, but also how to use the FZVis interactive compression evaluation and development platform.

Section 7: Open Discussion, Hands-On Q&A, and Closing Remarks (5 min.) RU

Interactive discussion with all presenters; feedback on attendee workflows, data reduction questions, and next-generation research opportunities.

Updated Slides

https://go.osu.edu/vis26



Closing

Thanks for participating in this tutorial

You can access/download the slides:

https://go.osu.edu/vis26